

Applied Machine Learning Lab

B.Tech Semester 4

Experiment 12

Unsupervised Pattern Recognition using K-Means Clustering on 2D Data

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SAP ID: 590016154

Name: Vidisha

AIM

To apply K-Means clustering on a synthetic 2D dataset, analyze the effect of different values of K, and determine the optimal number of clusters using evaluation metrics.

THEORY

K-Means is an unsupervised learning algorithm used to partition data into clusters by minimizing WCSS.

Steps include initializing centroids, assigning points, updating centroids and repeating until convergence.

Evaluation metrics include WCSS, Silhouette Score, Davies-Bouldin Index and Calinski-Harabasz Index.

Commands/Code Used

```
# Import libraries

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns


from sklearn.cluster import KMeans

from sklearn.metrics import silhouette_score, davies_bouldin_score,
calinski_harabasz_score


# -----

# GENERATE / LOAD DATA

# -----

# If you have CSV, use:

# df = pd.read_csv("data.csv")

# X = df.values


# Otherwise create synthetic dataset

from sklearn.datasets import make_blobs


X, y_true = make_blobs(n_samples=300, centers=3, cluster_std=0.6, random_state=42)


# -----

# VISUALIZE DATA

# -----
```

```

plt.figure(figsize=(6,5))

plt.scatter(X[:,0], X[:,1], c=y_true, cmap='viridis')

plt.title("Synthetic Dataset (True Clusters)")

plt.xlabel("Feature 1")

plt.ylabel("Feature 2")

plt.show()


# -----

# ELBOW METHOD (WCSS)

# -----

wcss = []

K_range = range(2, 11)


for k in K_range:

    kmeans = KMeans(n_clusters=k, random_state=42)

    kmeans.fit(X)

    wcss.append(kmeans.inertia_)


plt.plot(K_range, wcss, marker='o')

plt.title("Elbow Method (WCSS)")

plt.xlabel("Number of Clusters (K)")

plt.ylabel("WCSS")

plt.show()


# -----

# METRICS

```

```
# -----  
  
sil_scores = []  
db_scores = []  
ch_scores = []  
  
for k in K_range:  
    kmeans = KMeans(n_clusters=k, random_state=42)  
    labels = kmeans.fit_predict(X)  
  
    sil_scores.append(silhouette_score(X, labels))  
    db_scores.append(davies_bouldin_score(X, labels))  
    ch_scores.append(calinski_harabasz_score(X, labels))  
  
# Plot metrics  
plt.figure(figsize=(12,4))  
  
plt.subplot(1,3,1)  
plt.plot(K_range, sil_scores, marker='o')  
plt.title("Silhouette Score")  
  
plt.subplot(1,3,2)  
plt.plot(K_range, db_scores, marker='o')  
plt.title("Davies-Bouldin Index")  
  
plt.subplot(1,3,3)  
plt.plot(K_range, ch_scores, marker='o')
```

```
plt.title("Calinski-Harabasz Index")

plt.show()

# -----
# FINAL CLUSTERING (K=3)
# -----

kmeans = KMeans(n_clusters=3, random_state=42)
labels = kmeans.fit_predict(X)
centroids = kmeans.cluster_centers_

plt.figure(figsize=(6,5))
plt.scatter(X[:,0], X[:,1], c=labels, cmap='viridis')
plt.scatter(centroids[:,0], centroids[:,1], c='red', marker='X', s=200)
plt.title("Final Clustering (K=3)")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")

plt.show()
```

RESULTS

Synthetic dataset visualized.
Elbow method and evaluation metrics plotted.
Clustering results for different K values analyzed.
Final clustering obtained for optimal K=3.

CONCLUSION

K-Means successfully identified cluster structure. K=3 was optimal based on all evaluation metrics.